

RAG Project — Full Pipeline

From setup to deployed product. Every phase, every step.

Setup — Complete ✓

Ollama installed

Free local AI brain. llama3.2 downloaded.

Python libraries

langchain, chromadb, pypdf, langchain-ollama

VS Code + folder

research_agent project folder ready



Phase 1 — RAG Pipeline (you are here)

1. Create pdfs/ folder + drop in documents

Any text-based files — research papers, reports, anything

2. Load all documents

PyPDFLoader reads every file, extracts raw text page by page

3. Split into chunks

~500 words each with 50-word overlap so sentences don't get cut

4. Convert chunks to numbers (embeddings)

Ollama converts each chunk into a vector — numbers that capture meaning

5. Save to ChromaDB

Stored on disk. Done once. Loads instantly every time after that

6. Ask a question

Your question is embedded, ChromaDB finds closest chunks, Ollama answers

Checkpoint: RAG answers your questions



Evaluation — Is it actually correct?

Write test questions

10 Q&As; where you already know the right answer

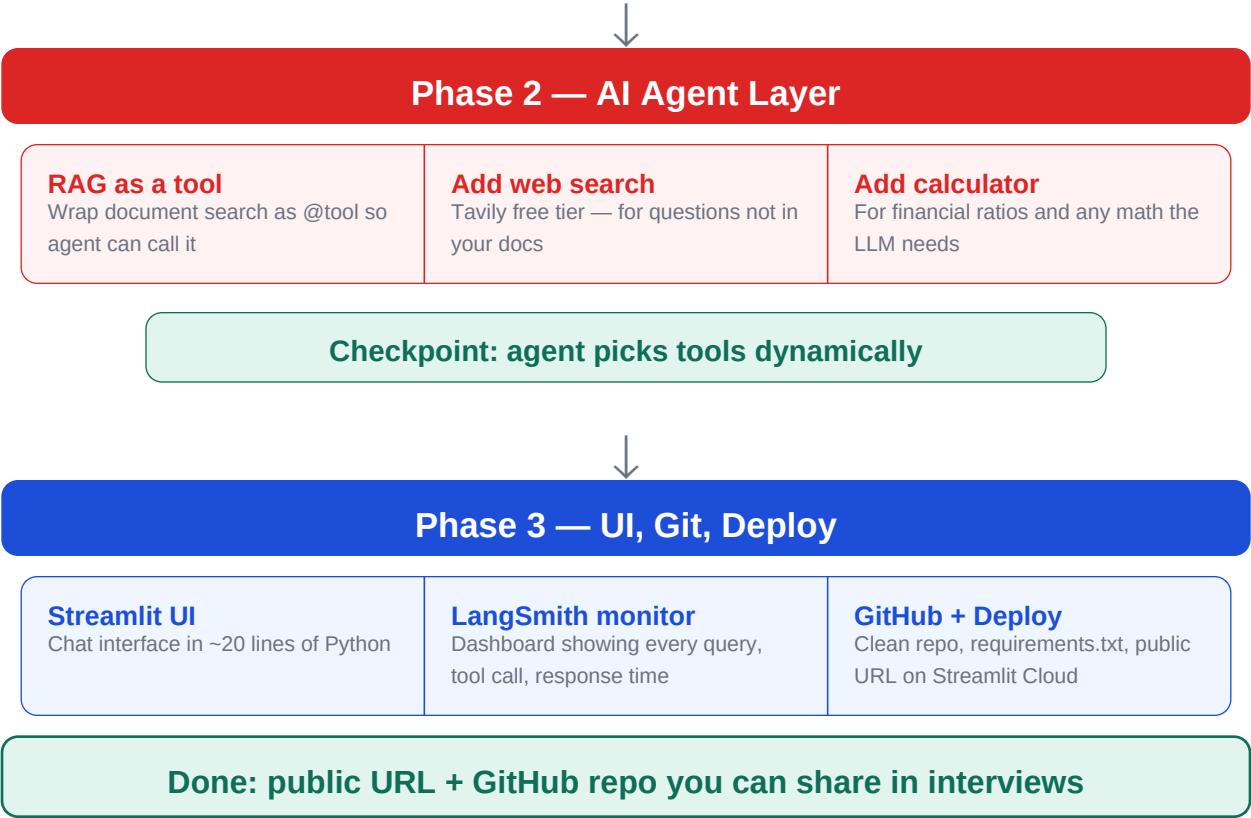
Run RAGAS scorer

Measures faithfulness and relevance automatically

Fix what is wrong

Tune chunk size, prompt, or k value until scores improve

Do not move to Phase 2 until RAGAS faithfulness score is above 0.7



What lives where

Your machine only	Goes to GitHub
Your PDF documents	Python scripts (ask_papers.py, agent.py, app.py)
ChromaDB database	requirements.txt
llama3.2 AI model (2GB)	README.md
Any API keys or secrets	Project folder structure